Demonstration of Proposed Features

Date	15 March 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Forensic Analysis Platform
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	**2D & 3D Video Surveillance Reconstruction**	OpenCV frame analysis detecting suspects, vehicles, weapons, 2D bounding boxes, and interactive 3D spatial trajectory room view.	Implemented	Yes	Demonstrated on 7 CCTV cases (retail shoplifting, traffic intersection, ATM assault, carjacking).
2	**Multilingual Forensic Audio Analysis**	OpenAI Whisper speech-to-text pipeline transcribing witness audio in English, Hindi, Telugu, and French with acoustic timeline extraction.	Implemented	Yes	Demonstrated on 911 distress calls, burglar alarms, and robbery witness depositions.
3	**Image Forensic Analysis & 3D Spatial Room View**	RT-DETR object detection, scene classification, bounding box overlays, and 3D wireframe room projection of evidence locations.	Implemented	Yes	High-resolution detection of suspects, firearms, and urban objects with confidence metrics.
4	**Second-by-Second Event Timeline**	Chronological crime event timeline mapping timestamps to security threats, motion breaches, and suspect actions with 1-click playback jump.	Implemented	Yes	Interactive timeline synchronized across video and audio media feeds.
5	**Interactive Multimodal AI Chatbot**	Context-aware forensic investigation assistant answering targeted queries regarding perpetrators, vehicles, and timing using Groq Llama-3.	Implemented	Yes	Session-grounded investigation assistant with direct timeline and evidence context retrieval.
6	**Automated Forensic PDF Report Generation**	One-click export of court-ready forensic incident reports containing case metadata, timeline, evidence logs, and confidence metrics.	Implemented	Yes	Court-admissible standardized PDF generated dynamically with ReportLab.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	**100%**